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Key Points:
- GICs measured at a low-latitude substation during geomagnetic storms are analyzed and simulated
- The first attempt to model the GICs at a low-latitude substation during storms by using the physical-based model
- The simulations capture the main active periods of GICs with comparable intensity to observations

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Measurements and Simulations of the Geomagnetically Induced Currents in Low-Latitude Power Networks During Geomagnetic Storms

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Abstract The issue of geomagnetically induced currents (GICs) on large distance ground-based conductive systems is a global space weather concern nowadays. The low-latitude power grids are becoming more vulnerable to GICs hazards due to their increasing scale and degree of interconnection. Thus, GICs measurement and modeling in these systems is of great importance. In this study, we present an analysis of the GICs measurement at a Chinese low-latitude substation during geomagnetic storms. The results show a rough positive correlation between the magnitude of large GIC spikes and $\mathrm{d}B_H/\mathrm{d}t$ spikes. We then built a physical-based model to simulate the GICs at the low-latitude substation during storms. It was found that the simulations capture the main active periods of the GICs during the storms with comparative strength of the measurement. This differs from the previous simulation results for high-latitude regions conducted by Welling et al. (2017; https://doi.org/10.1002/2016SW001505). Furthermore, the event analysis method is applied to evaluate the model performance. The results indicate that the physical-based model is more applicable than the persistence model in the prediction of GICs at low-latitude power grids during storms.

1. Introduction

Due to the interaction between solar wind and the Earth's magnetosphere, the time-varying external geomagnetic field induces an electric field on the ground, driving currents in large distance conductive ground-based systems such as power grids, pipelines, and railway systems (Boteler & Pirjola, 2017; Liu et al., 2016). The currents are known as geomagnetically induced currents (GICs). When the geomagnetic field varies drastically, GICs can grow large enough to disrupt the operation of the systems. For the power grids, large GICs can cause transformer malfunctions, permanent equipment damage, electric blackouts or even a collapse of the whole system (Pirjola, 2002). This could be one of the most critical space weather threats to human life in our era. The GICs threat to power transmission systems was originally recognized in high-latitude regions, like Scandinavia, Canada, and the northern United States (Boteler, 2001; Lam et al., 2002; Pulkkinen et al., 2008), which are often exposed to strong geomagnetic variations. Although the geomagnetic variations at middle and low latitudes are much weaker than those at high latitudes, many recent studies show that large GICs can also disrupt the middle- and low-latitude systems during strong geomagnetic activities (e.g., Carter et al., 2015; Gaunt & Coetzee, 2007; Liu et al., 2009; Marshall et al., 2012). Especially, the increasing scale of power networks at these latitudes makes them more vulnerable to GICs hazards. Zheng et al. (2012) showed that during a large storm, GICs in Chinese planned ultrahigh-voltage (UHV) transmission lines (1,000 kV) would exceed 100 A, or even reach to 200 A at some substations. The GICs issue is a global space weather concern nowadays. In order to understand and mitigate the potential GICs impact on power grids, a series of research including GICs measurement, modeling, forecasting, and hazards assessment has been conducted in many middle- and low-latitude countries (Liu et al., 2009; Marshall et al., 2011; Matandirotya et al., 2015; Watari et al., 2009; Zhang et al., 2015, 2016).

The ultimate goal of GICs research is to predict GICs in the conductive systems providing actionable information to the system operators for mitigating or preventing the hazards of GICs. To achieve this goal, physical-based global magnetohydrodynamics (MHD) models, driven by the observed upstream solar wind and interplanetary magnetic field parameters, have been applied to model the geomagnetic field perturbations dB or their time derivatives dB/dt (Pulkkinen et al., 2007, 2010; Raeder et al., 2001; Weigel et al., 2003; Wintoft, 2005; Yu & Ridley, 2008). For example, Yu and Ridley (2008) calculated the ground-based magnetic perturbations and their time derivatives dB/dt at ∼150 ground-based magnetometers for a number of storm time intervals using the University of Michigan's global MHD code. Pulkkinen et al. (2013) compared the simulated dB/dt from five different numerical geospace models with observations at 12 ground-based magnetometers during six geomagnetic events. Welling et al. (2017) reanalyzed the results of Pulkkinen et al. (2013) to better understand the model's performance. To further evaluate the model's performance on predicting GICs, Zhang et al. (2012) used the global PPMLR (extension of the Lagrangian version of the piecewise parabolic method)-MHD model (Hu et al., 2007) with a plane wave model to compute the GICs flowing in a Finnish 400 kV transformer and directly compared the modeled GICs with the measurements.

All of the above studies focused on forecasting the geomagnetic perturbation dB, and its time derivative dB/dt or GICs at high-latitude regions. There was few research involved with the prediction of GICs flowing in middle- and low-latitude power grids by using physical-based models. For middle- and low-latitude regions, the interplanetary shocks and the ring current intensifications during geomagnetic storms are considered to be the main cause of large GICs. Zhang et al. (2015) simulated the GICs in two Chinese low-latitude high-voltage substations caused by the interplanetary shock that occurred on 17 March 2015 on the basis of the global PPMLR-MHD model which was driven by solar wind conditions observed by satellites. The model reproduced the recorded GICs quite well, indicating that global MHD simulations can provide a powerful tool for predicting low-latitude GICs related to sudden storm commencement (SSC) or sudden impulse (SI). Based on results of MHD simulations, they further assessed the GIC risk at low-latitude regions under an extreme interplanetary shock event (Zhang et al., 2016). However, prediction of the GICs in these regions during geomagnetic storms by using physical-based global models has not been done so far. Since GIC signals are much more complicated and prolonged during geomagnetic storms than those during SSCs or SIs which are caused by interplanetary shocks, the prediction of the GICs during storms at these power grids would be much more challenging.

GICs with large amplitudes have been recorded in Chinese middle- and low-latitude power grids. Abnormal transformer noise (heavy buzzing sounds) associated with large GICs was observed in the Heihe power grid. During a severe geomagnetic storm on 9–10 November 2004, a GICs peak value of 75.5 A was recorded at the neutral point of No. 1 transformer at the Ling'ao nuclear power plant (Liu et al., 2009). Nowadays, China takes the GICs issue quite seriously and a multipoint GIC-monitoring program has been conducted since the beginning of 2015. More than one year's GICs data at five middle- and low-latitude substations were collected. The recorded GICs at the five monitoring points during this program are at a relative weak level (lower than 5 A) and did not disturb the operation of the power systems. But, analyzing these data is still important. On one hand, it can shed light on the signal characteristics of GICs during storms at middle- and low-latitude power grids. Although recent work involving GICs in these regions has been conducted, GICs measurement data are still scarce or rarely made public. GICs signal characteristics at these regions during geomagnetic storms has not been well analyzed and understood. For example, it is known that the time derivative of the horizontal geomagnetic field is highly related to the GICs for high-latitude regions (Viljanen et al., 2001), while Watari et al. (2009) reported that GICs have better correlation with geomagnetic field variations rather than with time derivatives of the geomagnetic field by using GICs measurements in a middle-latitude substation. The recent GICs data measured at the Chinese power grids will help to understand the GICs characteristic in these regions. On the other hand, if the measured GICs data could be used to estimate the performance of the physical-based GICs forecast model for middle- and low-latitude power grids, it would be a step forward for GICs forecast in middle- and low-latitude regions during geomagnetic storms.

In this study, we will present the GIC measurements at the Huangmeishan 220 kV substation recorded during storms and investigate the relationship between large GIC spikes and the time derivative of the horizontal geomagnetic field perturbations. We will then build a physical-based GICs forecast model for this substation to predict the GICs during storms. The simulated and measured GICs will be compared to estimate the model's performance. This would be the first attempt to model GICs at low-latitude power grids during storms by using the physical-based model. The structure of the paper is as follows. In section 2, we present the GICs recorded at the Huangmeishan 220 kV substation during three geomagnetic storms and the data analysis result. In section 3, we describe the methodology of building the physical-based GICs forecast model. Section 4 reports the simulation results and evaluation efforts. Finally, we discuss and summarize our study in sections 5 and 6.

2. GICs Measurements and Analysis

Considering the rapidly increasing scale and degree of interconnection of the high-voltage power networks, GICs problem has been taken seriously in China in recent years. Supported by the National High Technology Research and Development Program, GICs data were collected at five substations from the beginning of 2015 for more than 1 year. In this study, the GICs data measured at the Huangmeishan substation (27.37°N, 119.53°E in geographic coordinates, 16.85°N, 169.86°W in geomagnetic coordinates) during three representative geomagnetic storms of different intensities was presented and analyzed. Using the minimum Dst value as a criterion, Loewe and Prölss (1997) classified storms into five categories: weak (≤ -30 nT), moderate (≤ -50 nT), strong(≤ -100 nT), severe(≤ -200 nT), and great(≤ -350 nT). Because large GICs rarely appear in weak storms and no great storms occurred during the observation time period, we selected three storms of moderate, strong and severe magnitude for this study. The storms took place from 14–15 December 2015, 6–8 March 2016, and 17–18 March 2015 with their minimum SYM-H indices of −60 nT, −110 nT, and −232 nT, respectively. The SYM-H index could be considered as high-resolution Dst index (Wanliss & Showalter, 2006). Figure 1 shows the SYM-H index for each storm and the corresponding GICs data recorded at the Huangmeishan 220 kV substation. The three vertical red dashed lines in each panel indicate the start of each storm's initial phase, main phase, and recovery phase, respectively. The horizontal gray dashed lines in each GICs panel indicate the ±1 A levels of GICs.

The GICs monitor devices acquire the GICs signal at the transformer neutral point, as well as the vibration and noise signal at the transformer. With frequency of 0.0001–0.01 Hz, GICs can be considered as quasi-DC currents. In order to extract the GICs signal, the filtering algorithm and wavelet analysis method have been used to remove the vibration and noise signal (Wang et al., 2009). The magnitude of the GICs signal measured at the Huangmeishan 220 kV substation in this study is relatively weak, it is comparable to the measurements of GICs at the Memanbetsu substation in Hokkaido, Japan, from 2006 to 2007 (Watari et al., 2009). In their study, the maximum GICs range was between 0.66 and 3.85 A during geomagnetic storms. The GICs recorded at the Huangmeishan substation during these three storms have a maximum value less than 4 A; in reality that is not strong enough to disrupt the operation of the transformers. However, analysis and modeling of these data are still of great significance in understanding the features of GICs in these regions and evaluating the performance of the prediction models or assessing risk under extreme space weather conditions. The reason for the weak GICs measured in the Huangmeishan 220 kV substation will be discussed in section 5.

For the moderate storm (Figure 1a), the maximum and minimum GICs values are 1.09 and −1.48 A, respectively. For the strong storm (Figure 1b), the maximum and minimum values are 0.91 and −1.18 A, respectively. The severe storm (Figure 1c) is the strongest geomagnetic storm of the 24th solar cycle. The GICs signal for this storm is very special as the GICs are remarkably larger during the SSC than those during the storm main phase at low-latitude substations, which has been reported by Zhang et al. (2015). GICs caused by the SSC is 3.28 A at the Huangmeishan substation, while the maximum and minimum GICs values during the storm are 1.37 and −1.63 A, respectively. Typically, a storm lasts for several days. In order to investigate which phase the large GICs occur most frequently, we count the GIC spikes with an absolute value larger than 1 A during the three storms and sort them into different storm phases. The result is described in a histogram and are shown in Figure 2. There is a total of 24 GIC spikes with an absolute value larger than 1 A during these three storms. The counts are 1, 0, 20, and 3 for SSC, initial phase, main phase, and recovery phase, respectively. This indicates that the storm's main phase is the most dangerous period in which large GICs occur more frequently.

Furthermore, we investigate the relationship between GICs and the time derivative of the horizontal component of the geomagnetic field perturbations ($\mathrm{d}B_H/\mathrm{d}t$), $\mathrm{d}B_H/\mathrm{d}t = \sqrt{(\mathrm{d}B_x/\mathrm{d}t)^2 + (\mathrm{d}B_y/\mathrm{d}t)^2}$. The geomagnetic recordings from the nearest geomagnetic observatory to the substation, the Quanzhou geomagnetic

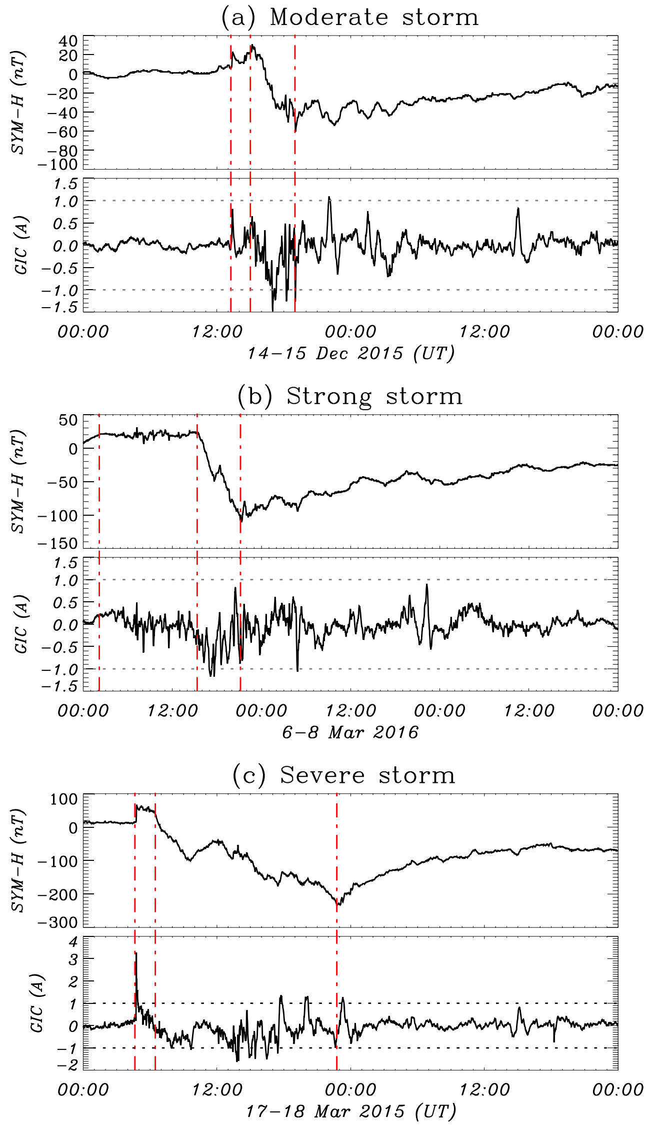

Figure 1. SYM-H index and the corresponding GICs measured at Huangmeishan 220 kV substation during (a) the moderate storm, (b) the strong storm, and (c) the severe storm. The three vertical red dashed lines in each panel indicate the start of the storm's initial phase, main phase, and recovery phase, respectively. The horizontal gray dashed lines in each GICs panel indicate the ±1 A levels of GICs.

observatory (24.98°N, 118.59°E in geographic coordinates), are used to calculate the time derivative of the geomagnetic field perturbations at the power network. It is ~280 km from the Quanzhou geomagnetic observatory to the Huangmeishan substation. In GIC studies, the measurements of the geomagnetic variations obtained sufficiently close to the substation are always used to represent the geomagnetic variations at the substation. The "sufficiently close" has been found to be of the order of 200 km for high-latitude region (Pulkkinen et al., 2006; Viljanen et al., 2004). The range can be larger for the low-latitude region due to the large scale of the geospace current systems which produce the geomagnetic perturbations in the low-latitude region. In this study, the geomagnetic variations measured at the Quanzhou geomagnetic observatory are used to represent the geomagnetic variations at the substation. Here we still focus on the large GIC spikes with an absolute value larger than 1 A. Figure 3 shows the north component and east component of the geomagnetic perturbations (dB_x, dB_y) recorded by the Quanzhou geomagnetic observatory, horizontal component of the time derivative of the geomagnetic perturbations dB_H/dt, and GICs measurements at the Huangmeishan 220 kV substation for the three

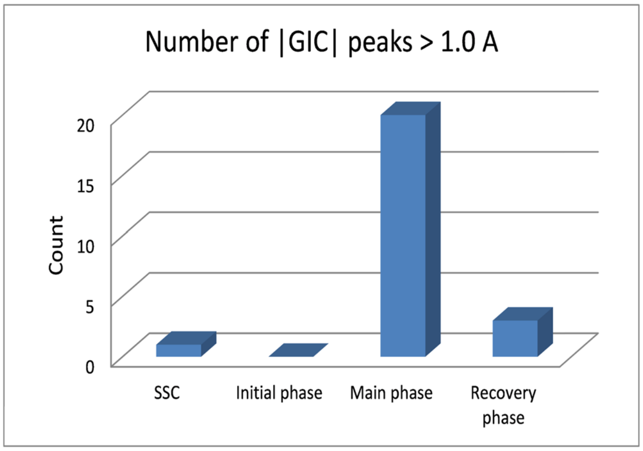

Figure 2. Counts of GIC spikes with absolute value larger than 1 for different storm phases during the three investigated storms in this study.

storms. The blue vertical dashed lines in the bottom two panels indicate the occurrences of the GIC spike with an absolute value larger than 1 A. The horizontal gray dashed lines in each GICs panel indicate the ±1 A levels of GICs. The figure shows that nearly all large GICs spikes correspond to spikes of dB_H/dt, which demonstrates clearly the close relationship between GICs and dB_H/dt. This is consistent with the study of Viljanen et al. (2001). From the figure, we can also see that the severe storm (Figure 3c) has the most GIC spikes, this is to be expected. What was unexpected is the number of large GIC spikes during the strong storm (Figure 3b) being less than those during the moderate storm (Figure 3a). This could attribute to the greater number of large dB_H/dt spikes during the moderate storm than those during the strong storm.

To investigate the relationship between the magnitude of GICs and dB_H/dt, we plot the amplitude of the large GIC spikes and the magnitude of their corresponding dB_H/dt during the three storms in Figure 4. The cross-correlation (CC) coefficient is labeled on the figure. The general trend is that the stronger the dB_H/dt is, the stronger the GICs spike is. The largest GICs spike during these three storms is 3.28 A, the data point is located at the top right corner of Figure 4. It was caused by the SSC during the severe storm. Regardless of this special point, the other GICs spikes occurred during storm main phases and recovery phases. If we focus on these data points, we notice that they are quite dispersed. They do not follow a simple linear relationship. The value of dB_H/dt which correspond to one GICs value can vary in quite a large range. This is understandable, due to the fact that the GICs flowing in the power networks are influenced by many complicated factors including geoelectric fields, topology, and electrical parameters of the power network, while the geoelectric fields depend on the direction and variation frequency of the geomagnetic perturbations as well as the Earth's conductivity. Obviously, the dB_H/dt cannot represent so many factors.

The above analysis indicates that though there is a rough positive relation between dB_H/dt and GIC, we should also keep in mind that the dB_H/dt cannot be equivalent to GICs completely. Their intensity relationship is complicated during geomagnetic storms in low latitudes.

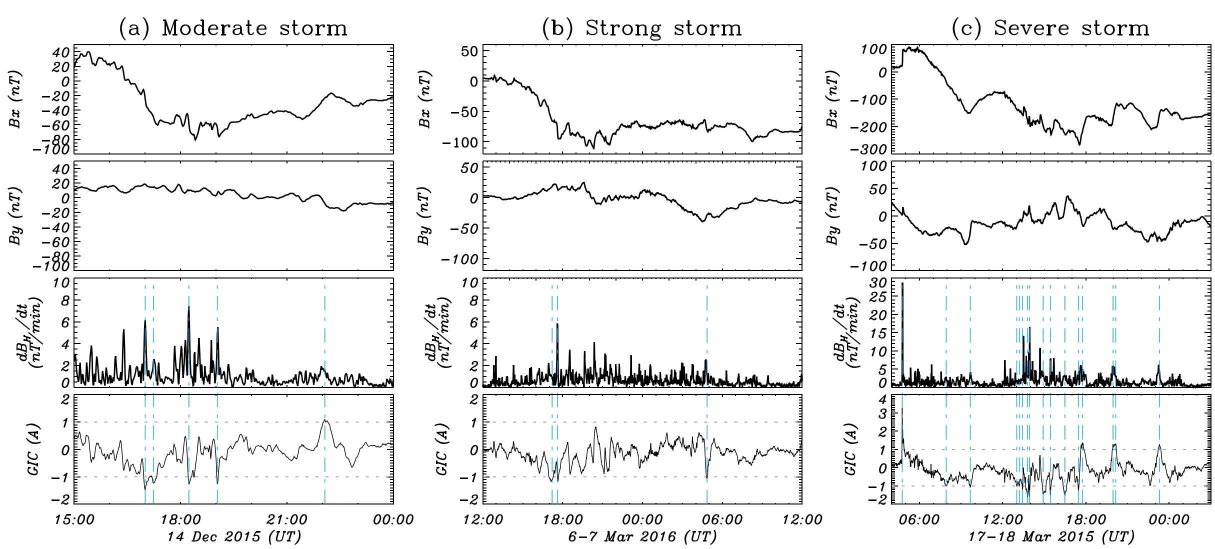

Figure 3. The X component and Y component of the geomagnetic perturbations (dB_x, dB_y) recorded by the Quanzhou geomagnetic observatory, the time derivative of the horizontal component of the geomagnetic perturbations dB_H/dt, and GICs measurements at the Huangmeishan 220 kV substation for (a) the moderate storm, (b) the strong storm, and (c) the severe storm. The blue vertical dashed lines in the bottom two panels indicate the occurrences of the GIC spike with an absolute value larger than 1 A. The horizontal gray dashed lines in each GICs panel indicate the ±1 A levels of GICs.

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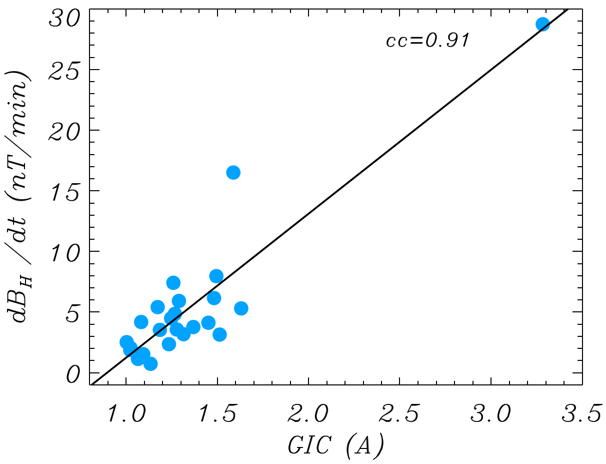

Figure 4. The relationship of the amplitude of the large GIC spike and the magnitude of their corresponding dB_H/dt, the cross-correlation coefficient is labeled on the figure.

3. Simulation Methodology

We will simulate the GICs at low-latitude power grids during the three geomagnetic storms. The simulation method is described in this section. First, we run the Space Weather Modeling Framework (SWMF) to calculate the geomagnetic perturbations at the substation with the in situ solar wind and interplanetary magnetic field data as input. Subsequently, we calculate the geoelectric fields at the substation by using the simulated geomagnetic perturbations and the Earth's conductivity model. Lastly, the GICs are calculated based on the geoelectric fields and the substation GICs coefficients.

3.1. The SWMF and Driving Conditions

The simulation is carried out using the SWMF developed at the University of Michigan (Tóth et al., 2005, 2012). It is a high-performance physics-based flexible software framework designed for space weather simulations. It divides the complex space physics systems ranging from the surface of the Sun to the surface of the Earth into multiple coupled physics domains. The domains used in this study are the Global Magnetosphere (GM), the Inner Magnetosphere (IM), and the Ionosphere Electrodynamics (IE). The GM domain is represented by the Block-Adaptive Tree Solar wind Roe-type Upwind Scheme (BATS-R-US) code which solves the equations of ideal MHD (Powell et al., 1999). The Ridley Ionosphere Model (RIM) (Ridley et al., 2004) is used for the IE domain. The Rice Convection Model (RCM) which simulates the ring current dynamics is used for the IM domain (Toffoletto et al., 2003; Wolf et al., 1982). In this study, the model is driven by the measured solar wind and interplanetary magnetic field (IMF) parameters as shown in Figure 5. The solar wind and IMF conditions for the moderate storm (Figure 5a) and the strong storm (Figure 5b) were downloaded from the NASA OMNI database (http://omniweb.gsfc.nasa.gov/ow.html). For the severe storm (Figure 5c), the

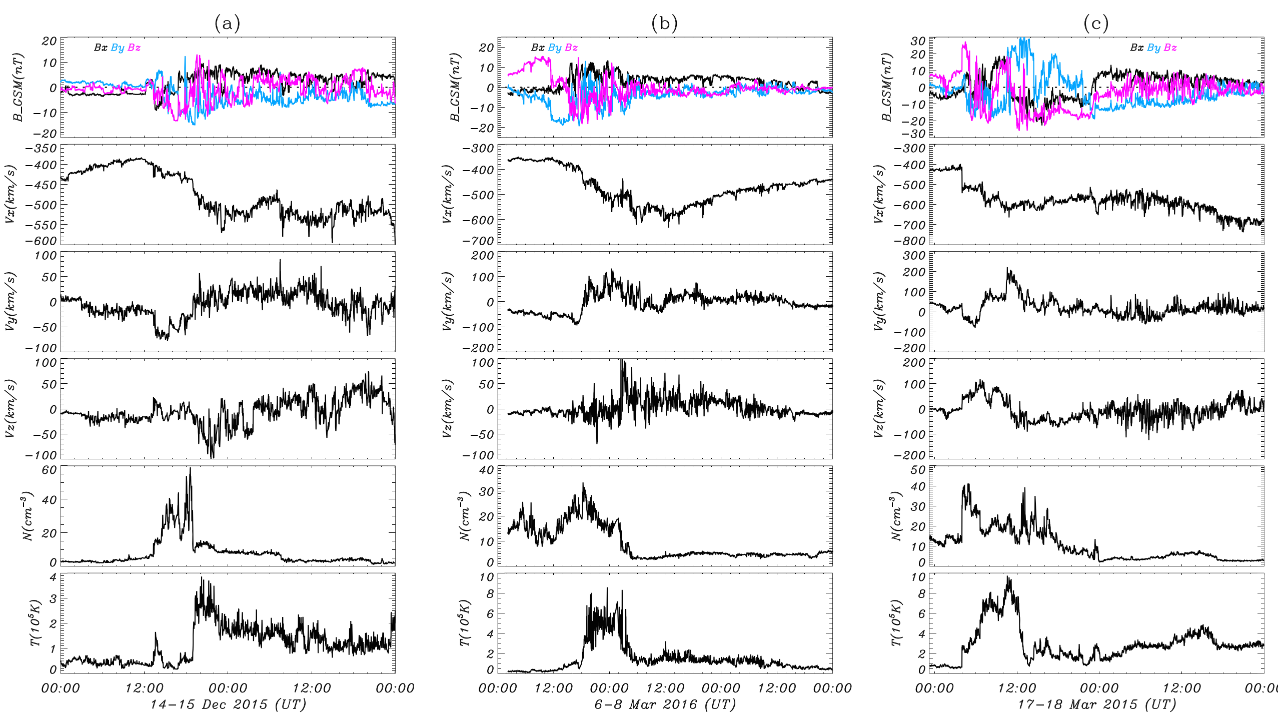

Figure 5. Solar wind and IMF observations during (a) the moderate storm occurred during 14-15 December 2015, (b) the strong strom occurred during 6-8 March 2016, and (c) the severe storm during 17-18 March 2015. From top to bottom, each plot shows three components of the IMF (B_x, B_y, and B_z), three components of the solar wind velocity (V_x, V_y, and Vz), the solar wind density (N), and the temperature (T).

Table 1
Ground Conductivity Model

Depth (km)	Conductivity (S/m)
0–30	0.005
30–90	0.013
90–150	0.04
>150	0.3

solar wind and IMF data were obtained from the WIND satellite, which has been time shifted to the nose of the bow shock in the figure. The data were downloaded from the CDAweb of Goddard Space Flight Center (https://cdaweb.sci.gsfc.nasa.gov/index.html/). The WIND data are used because there are too many data gaps in OMNI database during this event. In simulations, all of these data are time shifted to the inflow boundary of the BATS-R-US code.

The geomagnetic perturbations are calculated by integrating the magnetospheric current systems, ionospheric current systems, and field-aligned current systems with the Biot-Savart's Law (Yu et al., 2010).

3.2. GIC Calculation

The geomagnetic perturbations at the Huangmeishan 220 kV substation calculated from the SWMF model is used to calculate the local geoelectric fields based on the plane wave assumption (Cagniard, 1953). The geomagnetic perturbations are first transformed from geomagnetic coordinates to geographic coordinates. According to the plane wave assumption, the horizontal components of geoelectric field $E_{x,y}$ can be determined from the perpendicular geomagnetic field components $B_{y,\ x}$ in the frequency domain (Viljanen et al., 2004), as follows:

$$\widetilde{E}_x(\omega) = \frac{1}{\mu_0}\widetilde{Z}(\omega)\widetilde{B}_y(\omega) \tag{1}$$

$$\widetilde{E}_y(\omega) = -\frac{1}{\mu_0}\widetilde{Z}(\omega)\widetilde{B}_x(\omega) \tag{2}$$

where μ_0 is the vacuum permeability, Z is the surface impedance, depending on the Earth's conductivity structure. The subscripts x and y denote the north and east directions in geographic coordinates, respectively. For this study, a one-dimensional layered Earth's conductivity model is adopted for the area that the power grid is located in this study. This means the conductivity depends only on depth, with the lateral distribution of the conductivity at each layer being assumed to be uniform. The conductivity for each layer is shown in Table 1.

If we assume that the geoelectric fields are spatially uniform over the system region, GICs can be calculated straightforwardly in the time domain as follows:

$$GIC(t) = aE_x(t) + bE_y(t) \tag{3}$$

where a and b are constant coefficients, which depend on the electrical parameters and the topology of the power system. If we know the information about the network geometry, station coordinations, transformer resistances, transmission line resistances, and station earthing resistances, the two coefficients can be obtained theoretically (Lehtinen & Pirjola, 1985). They can also be obtained empirically if the geoelectric fields and the corresponding measured GICs are known (Pulkkinen et al., 2006; Wik et al., 2008; Zhang et al., 2012).

3.3. Coefficients Determination

In this study, we determine the coefficients a and b for the Huangmeishan 220 kV substation empirically. The ground conductivity model described in Table 1 and geomagnetic recordings from the Quanzhou geomagnetic observatory are used to derive the geoelectric fields. The coefficients a and b are then determined by the multiple regression analysis method based on the derived geoelectric fields and the recorded GICs at the substation during the three storm events. The data points during all the events and the fitting plane are shown in Figure 6. Based on the regression analysis, we get a=−152.08 A km/V, b=−103.68 A km/V for the substation.

Figure 7 shows the horizontal components of the observed geomagnetic disturbance at the Quanzhou geomagnetic observatory, and the calculated geoelectric fields, as well as the measured (black curve) and fitted (blue curve) GICs by using the coefficients a and b derived above at the substation during the three storm

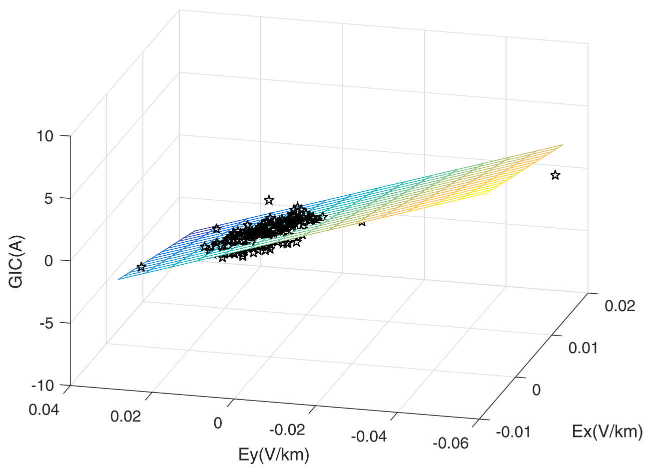

Figure 6. The fitting plane of the mutiple regression between GICs and geoelectric fields.

events. The CC coefficient and normalized root-mean-square (nRMS) difference between the observed and fitted GICs are labeled on the right of each GICs panel. The nRMS difference in this paper is defined as

$$nRMS = \sqrt{\frac{\langle (X_{model} - X_{obs})^2 \rangle}{\langle X_{obs}^2 \rangle}}$$

where (X_{model}) indicates the values calculated from a model, (X_{obs}) indicates the observation, and $\langle ... \rangle$ indicates arithmetic mean. The perfect value for the nRMS difference is zero, and less than 1 means good agreement between the trend of observation and modeled perturbation with some offset. The nRMS difference greater than 1 implies the modeled perturbation may miss the basic trend of the observation with a large deviation. It is shown that the fitted GICs curves match quite well with the measured data for all three storms. All the CC coefficients are larger than 0.85, and all the nRMS differences are about 0.5 for the three events.

In order to validate the coefficients a and b further, three other storm events during the GICs monitoring period are used to check if the fitted GICs could follow the observed GICs well. The results are shown in Figure 8, which is in the same format as Figure 7. The three storms occurred on 7–8 January 2015, 19–20 December 2015, and 30–31 December 2015, respectively. The CC coefficients between the fitted GICs and the observed GICs are 0.90, 0.89, and 0.88 for the three events. The nRMS differences are 0.42, 0.51, and 0.51, respectively. The high CC coefficients and the low nRMS differences indicate the reliability of the coefficients a and b.

4. Simulation Results and Evaluation

4.1. Simulation Results

According to the methodology described in section 3, we simulate GICs at the Huangmeishan 220 kV substation during the three storms with different magnitude, which was shown in Figure 1. Figure 9 shows

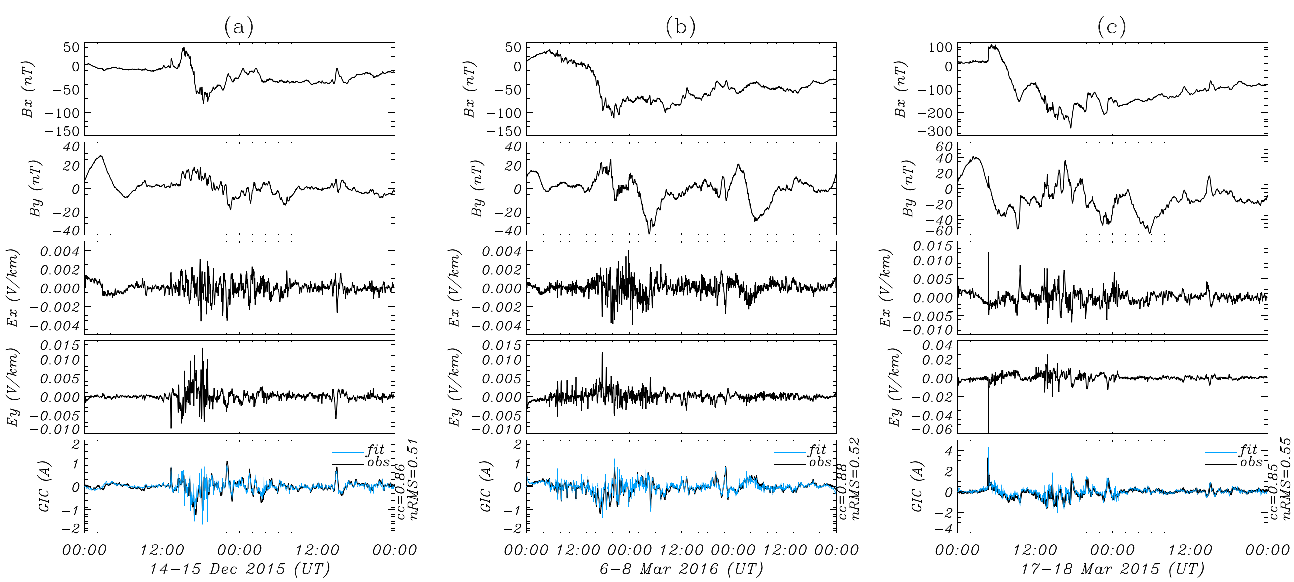

Figure 7. The horizontal components of the observed geomagnetic disturbance at the Quanzhou geomagnetic observatory (B_x, B_y), and the calculated geoelectric fields (E_x, E_y), as well as the measured (black curve) and fitted (blue curve) GICs at the substation during (a) the moderate storm, (b) the strong storm, and (c) the severe storm. The cross-correlation (CC) coefficient and nRMS difference between the observed and fitted GICs are labeled on the right of each GICs panel.

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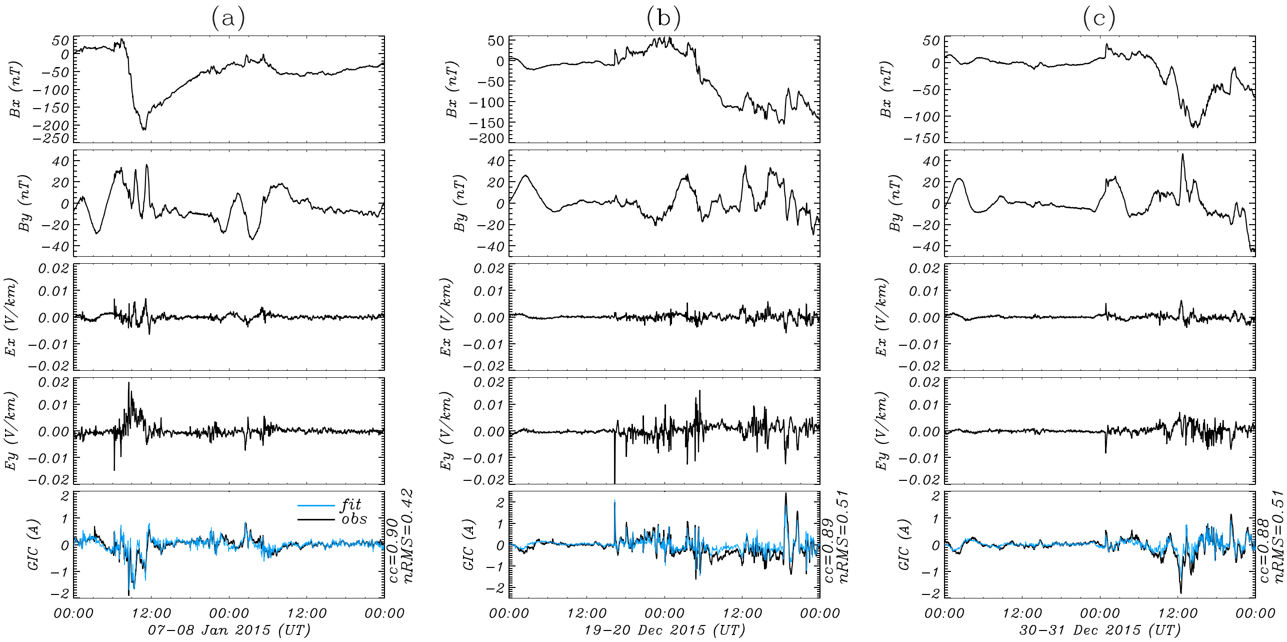

Figure 8. The horizontal components of the observed geomagnetic disturbance at the Quanzhou geomagnetic observatory (B_x, B_y), and the calculated geoelectric fields (E_x, E_y), as well as the measured (black curve) and fitted (blue curve) GICs at the substation during the storm occurred on (a) 7–8 January 2015, (b) 19–20 December 2015, and (c) 30–31 December 2015. The cross-correlation (CC) coefficient and nRMS difference between the observed and fitted GICs are labeled on the right of each GICs panel.

the simulated (red) and observed (black) x and y components of the horizontal geomagnetic variation B_x and B_y, the x and y components of the calculated horizontal geoelectric fields E_x and E_y, and the GICs. The CC coefficients and nRMS difference between the simulated and the observed B_x and B_y are labeled on the right

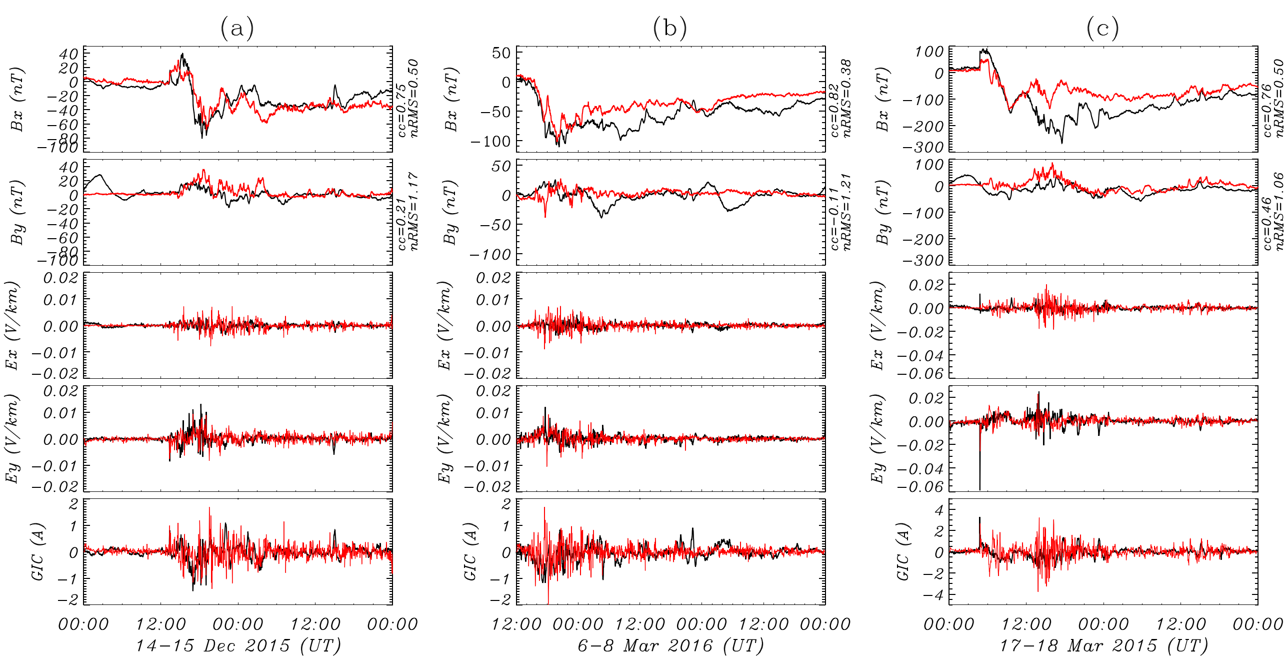

Figure 9. The simulation results (red) and observation (black) of the x and y components of the horizontal geomagnetic variations, the x and y components of horizontal geoelectric fields, and the GICs at the Huangmeishan 220 kV substation during (a) the moderate storm of 14–15 December 2015, (b) the strong storm of 6–8 March 2016, and (c) the severe storm of 17–18 March 2015.

of each panel. The model captures the perturbation of B_x very well with high CC coefficients and low nRMS differences for all events. All of the CC coefficients are higher than 0.75, and nRMS errors are equal to or less than 0.50. The highest CC coefficient of 0.82 and lowest nRMS difference of 0.38 appears in the strong storm (Figure 9b). The model could not reproduce the variations of B_y so well for all events. Among the three events, the highest CC coefficient for B_y is 0.46, which appears in the severe storm (Figure 9c). The worse prediction of B_y is probably attributed to an over-rotation of the ring current toward the dayside in the model (Yu et al., 2010). Nevertheless, the northward component of the geomagnetic perturbations B_x at low latitudes is the dominant component during storms. The variations of E_x, E_y are much more complicated than B_x and B_y. Generally speaking, simulations capture the main active periods of E_x, E_y for each storm. And the simulation captures the timing of major activities of GICs and they are at a comparable level of intensity as in the observations.

4.2. Evaluation

To evaluate the performance of the model in predicting GICs, we use the event-based analysis method to evaluate the ability of the model in reproducing GICs, as applied extensively in previous studies (Pulkkinen et al., 2007, 2013; Zhang et al., 2012). An event is identified when the absolute value of GICs exceeds a threshold $|GIC_{thres}|$ within a forecast window $0 \leq t \leq t_f$. The windows are moved over the time series in nonoverlapping segments. Events are recorded for both the observed and the simulated GICs for the given t_f and $|GIC_{thres}|$. By comparing the threshold crossing for both observed and simulated GICs, the contingency table is built, which consists of numbers of correct hits (H), false alarms (F), missed events (M), and no events (N). As described by Pulkkinen et al. (2013), the set of $[H, F, M, N]$ is used to calculate three metrics: Probability of Detection (POD), Probability of False Detection (POFD), and Heidke Skill Score (HSS). POD is defined as $POD = H/(H + M)$, measuring the fraction of threshold crossings that are predicted correctly. It ranges from 0 to 1, with 1 being the perfect score. POFD is defined as $POFD = F/(F + N)$, which measures the fraction of no crossings that are incorrectly forecasted as crossings. It ranges from 0 to 1, with 0 being the perfect score. The metrics POD and POFD are usually used in pair, since a model predicting artificially large GICs tends to generate large H and large POD, while a model providing artificially low GICs will generate small F and small POFD. HSS measures the fraction of correctly predicted threshold crossings after eliminating the random correct predictions. The definition of HSS is $HSS = 2(HN - MF)/[(H + M)(M + N) + (H + F)(F+N)]$. It ranges from negative infinity to 1, with 1 being the perfect score, 0 indicating that the prediction is as good as random, and negative values representing that the prediction of model is worse than the random prediction.

In this study, we set the length of the analysis window t_f as 30 min. The maximum absolute value of GICs for the three storms are 1.48, 1.18, and 3.28 A, respectively. If the threshold $|GIC_{thres}|$ exceeds the observed maximum absolute value of GICs, then $H = M = 0$, meaning that $POD = 0/0$, which is not defined. If the maximum absolute values of GICs for both simulation and observation during an event are lower than the threshold ($H = M = F = 0$), HSS would be 0/0. In order to obtain a defined metrics, GICs thresholds $|GIC_{thres}|$ is chosen from 0.25 to 1.0 A with an increment of 0.25 A for the moderate and strong storms. The largest $|GIC_{thres}|$ is chosen as 2.5 A for the severe storm. As the maximum absolute GICs for this storm main phase is 1.68 A, POD keeps at 1 with increasing $|GIC_{thres}|$ when thresholds are larger than 1.75 A, which is caused by the correct prediction of GICs due to the SSC prior to the storm main phase. Figure 10 shows the POD, POFD, and HSS as a function of event threshold for the three storms from the model forecast (magenta), and from a persistence model forecast (blue). In the persistence model("nearest neighbor forecast" model), the observation for the current window is the forecast for the next forecast window.

Figure 10 shows that simulation provides higher POD and POFD than the"nearest neighbor forecast" model almost for all thresholds in the three storms. The POD from the SWMF simulation remains at a relatively high level as $|GIC_{thres}|$ increases, while POFD decreases rapidly to nearly zero (the perfect value of POFD) with increasing $|GIC_{thres}|$. This indicates that the physical-based model produces artificially larger signal amplitudes than in reality at low GIC levels; when the threshold increases, the physical-based model produces pretty good prediction with high POD and very low POFD. For the "nearest neighbor forecast" model, POD decreases rapidly with increasing $|GIC_{thres}|$, while POFD remains at a very low level, implying that the "nearest neighbor forecast" model predicts low GICs for large $|GIC_{thres}|$, which means it cannot predict large sudden GIC spikes.

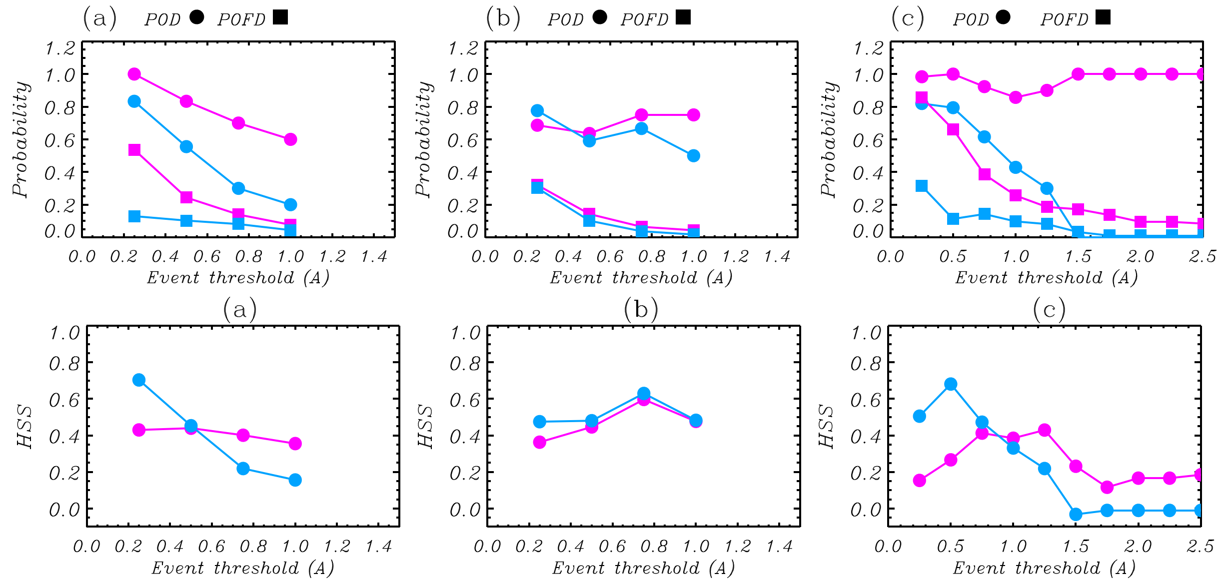

Figure 10. (top row) Probability of Detection (POD), Probability of False Detection (POFD), and (bottom row) Heidke Skill Score(HSS) as a function of event threshold for the physical-based model (magenta curves) and the "nearest neighbor forecast" model (blue curves) for (a) the moderate storm of 14–15 December 2015, (b) the strong storm of 6–8 March 2016, and (c) the severe storm of 17–18 March 2015.

The HSS for the simulation are larger than that of the the "nearest neighbor forecast" model when thresholds are larger than 0.5 A for the moderate storm of 14–15 December 2015. For the strong storm, the HSS values are very close to each other for all thresholds, varying between 0.4 and 0.63. For the severe storm, HSS from MHD simulations are higher than the "nearest neighbor forecast" model for thresholds larger than 1.0 A, with the highest HSS approaching 0.43 when the threshold is 1.25 A. When GICs > 1.5 A, HSS from the persistence model are less than 0, indicating that the model prediction is worse than the random forecast. While the HSS for the physical-based simulation are still larger than 0. The above results suggest that for weaker GICs cases (GICs < 1 A), the persistence model shows some advantage over the physical-based model. However, the physical-based model is more superior to the persistence model for larger GIC cases (GICs ≥ 1 A), indicating that it is more applicable for predicting the strong GICs perturbations. Because the large GICs at low latitudes are predominately associated with the variations of the ring current, we infer that this is because the dynamics of the ring current during strong storms is adequately described by the SWMF model. This is different from the results of Welling et al. (2017) who reanalyzed the ability of five geospace models including the SWMF model to predict dB/dt at high latitudes. Their results showed that all models tend to underestimate the observed dB/dt. At low latitudes, the geospace current systems responsible for GICs are different from that at high latitudes. The magnetopause currents and ring currents are the main sources of geomagnetic variations that cause GIC hazards to the networks at low latitudes, while at high latitudes, the main sources of large dB/dt or GICs are ionospheric electrojets. Although the SWMF model underpredicted the GIC related dB/dt at high latitudes especially during active periods, our results show that the SWMF model could reproduce the GICs at low latitudes during the main active period.

5. Discussion

According to the description in section 3.2, the intensity of GICs in power grids depend on the geomagnetic drivers, the ground conductivity, as well as the topology and electrical parameters of the network. The GICs measured at the Huangmeishan 220 kV substation during the investigated period are relatively weak, the maximum GICs value is 3.28 A during the severe storm. The largest GICs ever recorded in Chinese power networkswas that measured in the Ling'ao 500 kV substation (geographic latitude 23°, geomagnetic latitude 12°) in the Guangdong power network during a severe storm on 9–10 November 2004 (Liu et al., 2009). The magnitude of the storm is comparable to the severe storm on 17–18 March 2015 in this study. The measured GICs, time derivative of the horizontal component of the geomagnetic perturbations dB_H/dt, the north

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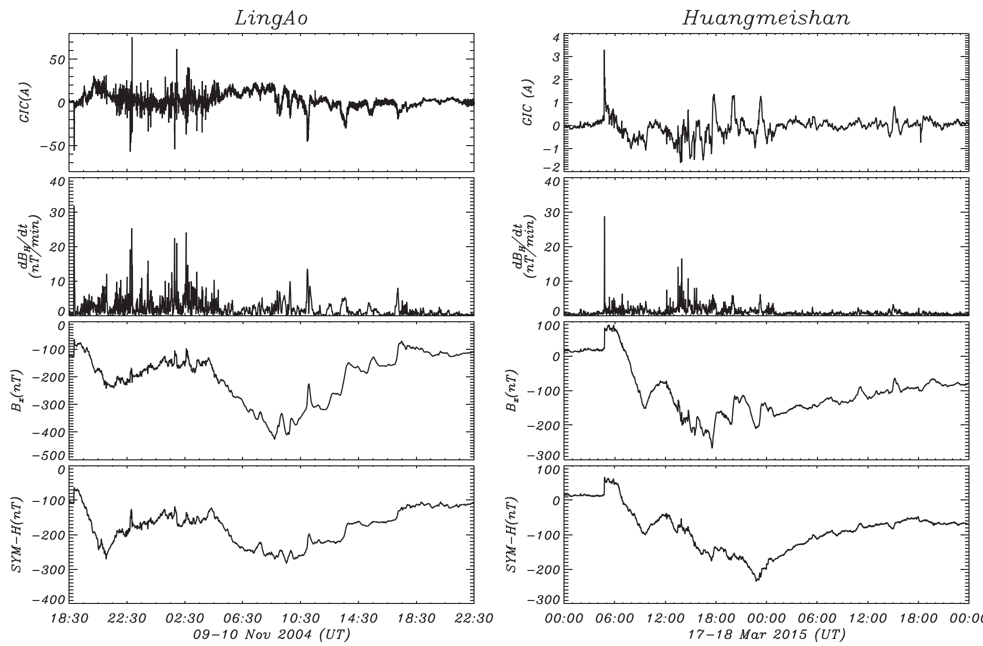

Figure 11. From top to bottom: (first row) The measured GICs, (second row) time derivative of the horizontal component of the geomagnetic perturbations dB_H/dt, (third row) the north component of the geomagnetic perturbation B_x at the substations, and (fourth row) the SYM-H index during the two events (left) Ling'ao and (right) Huangmeishan.

component of the geomagnetic perturbation B_x at the two substations, and the SYM-H index during the two storms are shown in Figure 11 to compare. At the Ling'ao substation, the maximum GICs value of 75.5 A was recorded at 22:50 UT on 9 November 2004, the corresponding value of dB_H/dt is 25 nT/min. For the event that occurred on 17–18 March 2015, the maximum GICs value recorded at the Huangmeishan 220 kV substation is 3.28 A at 04:44 UT on 17 March 2015, the corresponding dB_H/dt is 28 nT/min. The two

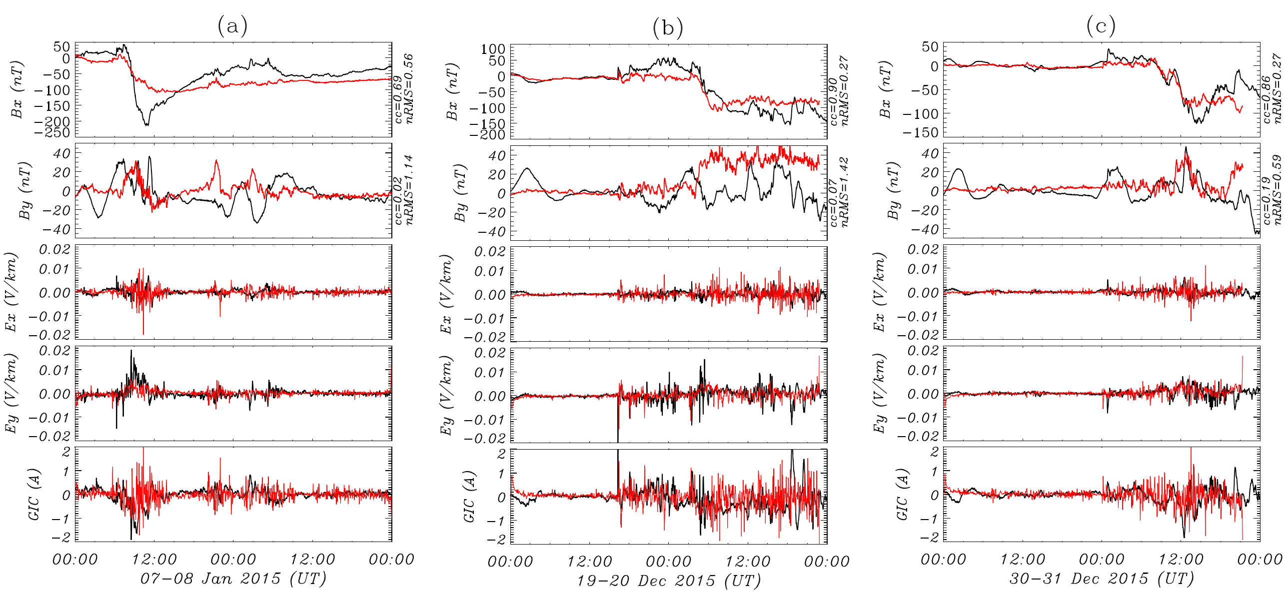

Figure 12. The simulation results (red) and observation (black) of the x and y components of the horizontal geomagnetic variations, the x and y components of horizontal geoelectric fields, and the GICs at the Huangmeishan 220 kV substation during the storm of (a) 7–8 January 2015, (b) 19–20 December 2015, and (c) 30–31 December 2015.

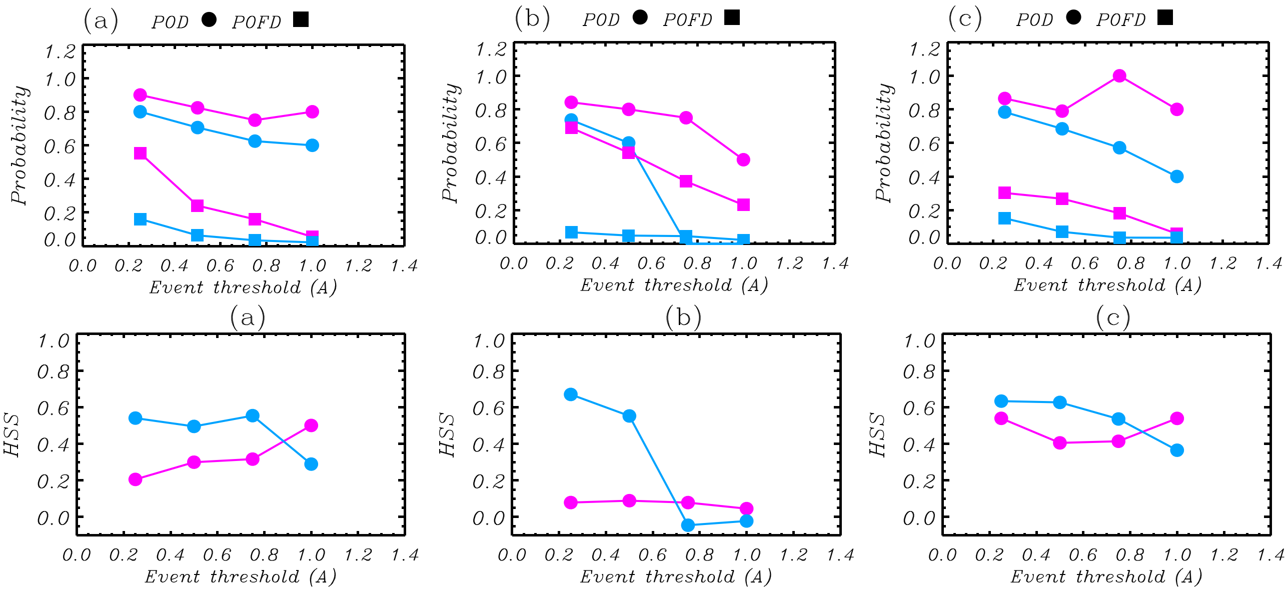

Figure 13. (top row) Probability of Detection (POD), Probability of False Detection (POFD), and (bottom row) Heidke Skill Score(HSS) as a function of event threshold for the physical-based model (magenta curves) and the "nearest neighbor forecast" model (blue curves) for the storm of (a) 7–8 January 2015, (b) 19–20 December 2015, and (c) 30–31 December 2015.

dB_H/dt values are very close, which indicates that the geomagnetic divers are not the reason for the weak GICs measured at the Huangmeishan 220 kV substation in this study. Further, the typical layered ground conductivity model for the Guangdong province of China is 0.0005S/m for 0–100 km, 0.01S/m for 100–150 km, 0.1S/m for 150–200 km, and 0.01S/m for a depth that is deeper than 200 km. This is much lower than the conductivity in the Huangmeishan substation which was shown in Table 1. Under the same geomagnetic perturbations level, ground with higher conductivity will induce lower geoelectric fields and vice versa. Therefore, the higher ground conductivity can be one of the reasons for the weak GICs measured at the Huangmeishan 220 kV substation. In addition, different conductors are used for the transmission lines with different voltage levels. The one conductor or double bundle conductor is used for the 220 kV transmission line, while a quad bundle conductor is usually applied for the 500 kV transmission line in China. The direct-current resistance per unit length of the double bundle conductor can be approximately an order higher than the quad bundle conductor. Thus, the high direct-current resistance of the transmission line should be another major reason for the weak GICs measured in the Huangmeishan 220 kV substation. Besides, the topology of the power network can also affect the magnitude of the GICs.

In section 4, we show the simulation results and evaluation for the three storm events with different intensities. The results indicate that the simulation could capture the main features of the GICs. The physical-based model is more superior to the persistence model for larger GIC cases. Besides these three storm events, we also simulated the GICs during the three other storm events which have been used to validate the coefficients a and b adequately (Figure 8). The simulation results are shown in Figure 12, which is in the same format as Figure 9. And as was done in section 4.2, we also calculated the POD, POFD, and HSS for the physical-based model and the"nearest neighbor forecast" model for these three events. The results are shown in Figure 13. The figures indicate that the simulation and evaluation results are consistent with the results in section 4, which further support the conclusion made in section 4.

6. Summary

In this study, we presented and analyzed the GICs data recorded at a Chinese low-latitude substation during three storms with different intensities. The results showed that the storm main phase is the most dangerous phase in which large GICs spikes occur most frequently. The GICs spikes are closely related to the time derivative of the horizontal component of the geomagnetic perturbations in the low-latitude power grid. But the relationship between the intensity of GIC spikes and the dB_H/dt spikes during storms is complicated. These results showed that dB_H/dt can be a rough GIC activity indicator at low latitudes, but it reminded us that the dB_H/dt cannot completely replace GICs.

We simulated the GICs at the substation during the three storms by combining the SWMF model with a one-dimensional ground conductivity model. The simulation results indicated that the model captures the magnetic perturbation B_x very well, although B_y was not so well reproduced. Generally speaking, the SWMF-based simulations capture the fluctuation of the main active periods of GICs, and the strength of the simulated and observed GICs are comparative. This is different from the the model's performance at high latitudes, which also indicated the necessity to evaluate the performance of the model by different latitude regions. To evaluate the model's performance in predicting GICs, event-based analysis method is applied in this study. Results indicated that the physical-based prediction is more applicable than the persistence forecast model for the prediction of large GICs at low-latitude power grids during storms in practice. This work is the first attempt to model the GICs at low-latitude power grids during storms by using the physical-based model. More GICs observations will be collected and simulated to sufficiently evaluate the model's performance in the future.

Data Availability Statements

The GIC measurement data can be downloaded from Zenodo (https://10.5281/zenodo.3714269). The simulated geomagnetic perturbations data can be downloaded from Zenodo (https://10.5281/zenodo.1248238).

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